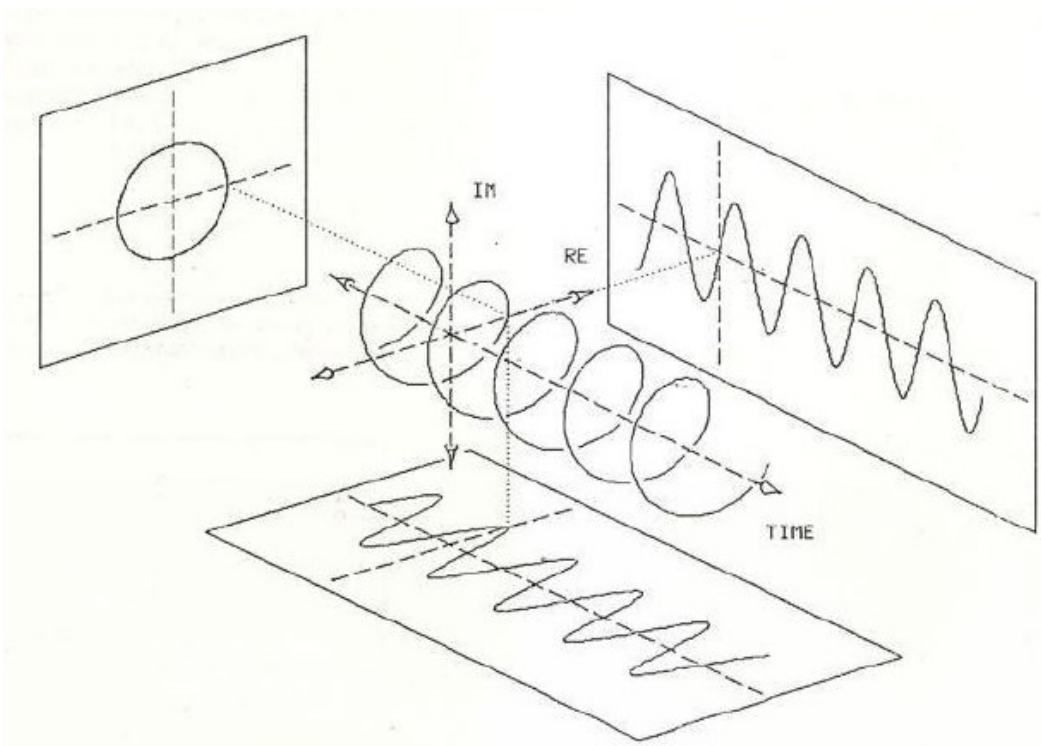
THE COMPLEX EXPONENTIAL

What does $e^{i\theta}$ mean?



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1 Introduction to the Complex Numbers

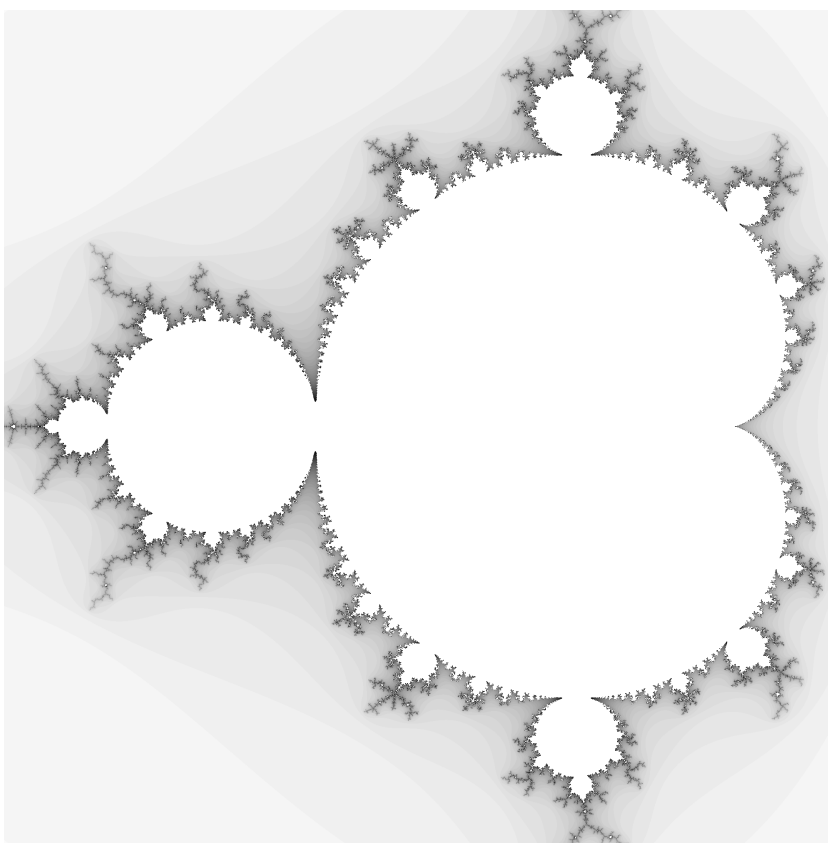
We learned about $\mathbf{i} = \sqrt{-1}$ when we were solving for the roots of polynomials. Numbers of the form $b\mathbf{i}$ are called **Imaginary Numbers**, and numbers of the form $a + b\mathbf{i}$ are called **Complex Numbers**. (Notice that the Imaginary Numbers and Real Numbers are subsets of the Complex Numbers). Operations using the complex numbers are fairly simple, just follow normal algebraic rules and use the definition of $\mathbf{i}$ when you need to.
For a complex numbers $z_0 = a + b\mathbf{i}$ and $z_1 = c + d\mathbf{i}$, and a real number p:

$$z_0 + z_1 = (a + b\mathbf{i}) + (c + d\mathbf{i}) = (a + c) + (b + d)\mathbf{i} \tag{1}$$

$$z_0 - z_1 = (a + b\mathbf{i}) - (c + d\mathbf{i}) = (a - c) + (b - d)\mathbf{i} \tag{2}$$

$$p \cdot z_0 = p(a + b\mathbf{i}) = pa + pb\mathbf{i} \tag{3}$$

The goal of this assignment is to give you an intuitive idea of what complex exponentials are, and why they are useful. At first glance, using an imaginary or complex number as an exponent seems meaningless. Most explanations use calculus or infinite series approximations, but I think that there is a very natural way to view complex exponentials through the lens of exponent laws and rotations. This turns out to be an extremely useful mathematical idea. It is used in Quantum Mechanics, electronic circuits, 3-D modeling, and many other branches of math and science. It is so convenient to work with, that the complex exponential replaces sine and cosine almost entirely in advanced physics.
Complex Numbers can also generate beautiful fractals like this one:



1.1 The Complex Plane

The complex plane is a graphical way to represent the set of complex numbers $\mathbb{C}$. We can take a complex number $z = a + b\mathbf{i}$ and split it into its real and imaginary parts.
The real part will become our x-coordinate:

$$\mathrm{Re}(z) = \mathrm{Re}(a + b\mathbf{i}) = a \tag{4}$$

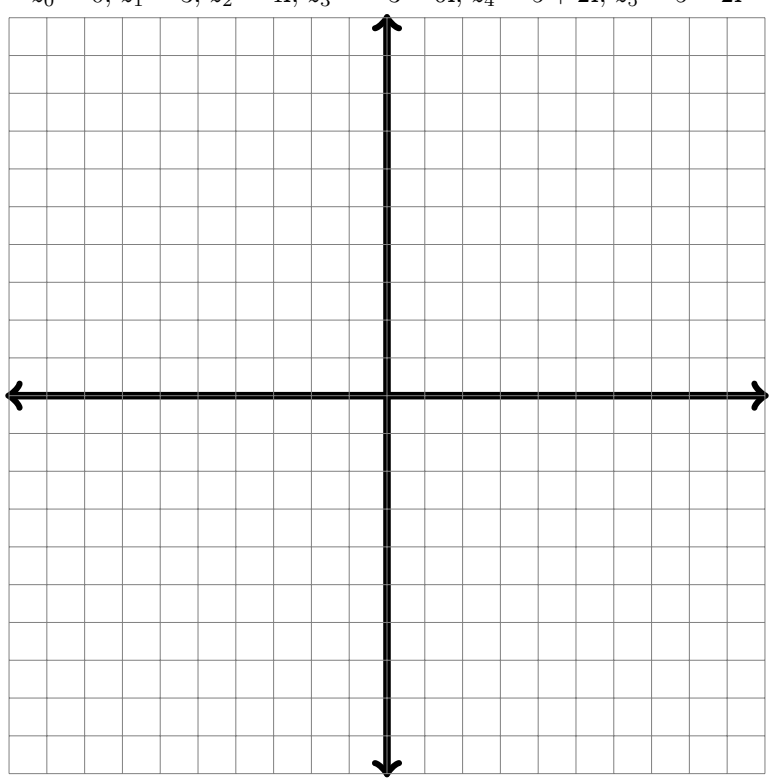
The imaginary part will become our y-coordinate:

$$\mathrm{Im}(z) = \mathrm{Im}(a + b\mathbf{i}) = b \tag{5}$$

Principle 1.1. To plot $z = a + b\mathbf{i}$ on the complex plane, we just plot the point (a, b)

1.1.1 Problem 1: Plot (and label) the following points on the complex plane

$z_0 = 0$, $z_1 = 3$, $z_2 = 4\mathbf{i}$, $z_3 = -3 - 6\mathbf{i}$, $z_4 = 5 + 2\mathbf{i}$, $z_5 = 5 - 2\mathbf{i}$



1.2 Complex Multiplication

In this section we will examine the properties of the complex numbers under multiplication. When we are working with the real numbers $\mathbb{R}$, multiplication can be thought of as an operation that magnifies, shrinks, or flips a length. $5 \cdot 2 = 10$ means that 10 is twice as long as 5 (or that 10 is 5 times the length of 2). $3 \cdot -1 = -3$ represents flipping a positive length of 3 backwards. We will see that the complex numbers share this property, and have an additional property of creating rotations.

Principle 1.2. Multiplying by a complex number can be seen as a stretch combined with a rotation

Let's actually multiply some complex numbers together. The crucial thing here is to remember to use the distributive property (FOIL). If we have two arbitrary complex numbers $z = a + b\mathbf{i}$ and $w = c + d\mathbf{i}$, then their product is:

$$z \cdot w = (a + b\mathbf{i}) \cdot (c + d\mathbf{i}) = ac + ad\mathbf{i} + bc\mathbf{i} + bd\mathbf{i}^2 \tag{6}$$

Using the fact that $\mathbf{i}^2 = -1$ and combining like terms,

$$z \cdot w = (ac - bd) + (ad + bc)\mathbf{i} \tag{7}$$

1.2.1 Problem 2: Use Equation 7 to multiply the following complex numbers

a) $\mathbf{i} \cdot (1 + \mathbf{i})$

b) $(1 + 2\mathbf{i}) \cdot (3 + \mathbf{i})$

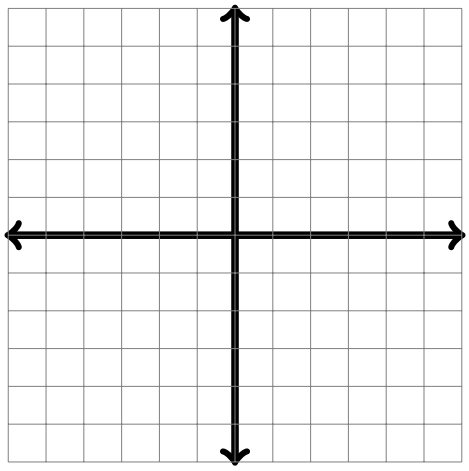
c) $(1 + \mathbf{i}) \cdot (1 - \mathbf{i})$

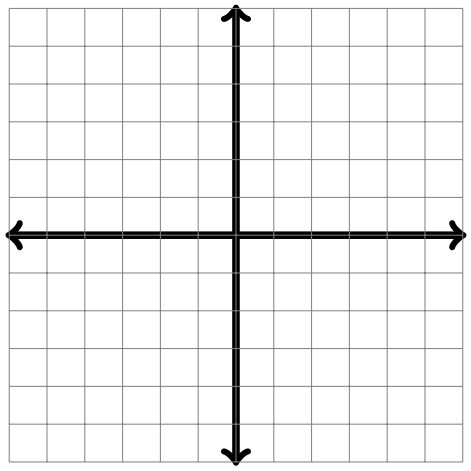
d) $(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\mathbf{i}) \cdot (\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\mathbf{i})$

1.2.2 Problem 3: Multiplication as Rotation

a) plot the following points on the complex plane below. Then, multiply each of them by $\mathbf{i}$ and plot those points on the second complex plane. Connect the dots to make a ;) face

$$3 + 3\mathbf{i},\ 2 + 3\mathbf{i},\ -2 + 3\mathbf{i},\ -3 + 3.5\mathbf{i},\ -6,\ -4 - 2\mathbf{i},\ -3\mathbf{i},\ 4 - 2\mathbf{i},\ 6$$



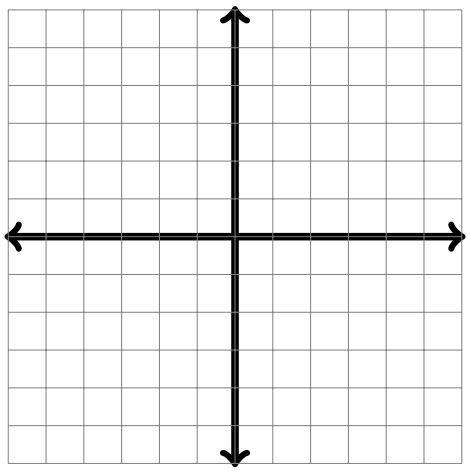


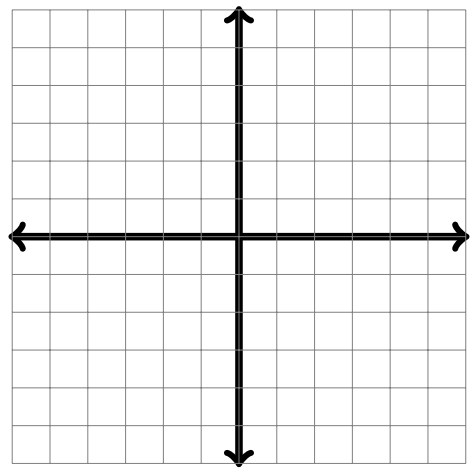
b) What geometric transformation relates the two pictures?

Principle 1.3. Multiplication by $\mathbf{i}$ rotates a point on the complex plane by _____ $^\circ$ or _____ radians counterclockwise about the origin

1.2.3 Problem 4: Multiplication as Scaling (Stretching)

a) plot the same set of points from question 3 on the complex plane below. Then, multiply each of them by $\frac{1}{2}$ and plot those points on the second complex plane. Connect the dots to make a ;) face





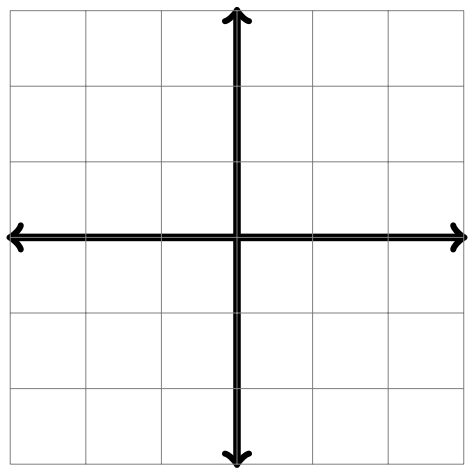
b) What geometric transformation relates the two pictures?

Principle 1.4. Multiplication by a real number magnifies/compresses a complex number's distance from the origin in the complex plane.

1.2.4 Problem 5: How Much Rotation? How Much Stretching?

We know that multiplying by $\mathbf{i}$ rotates a complex number by 90°, but what about other complex numbers? Different ones will rotate and stretch by different amounts. One way to visualize this rotation is by thinking of complex numbers as vectors (arrows) rather than singular points.

a) On the complex plane below, plot the complex number $z = 1 + 2\mathbf{i}$, then draw an arrow from the origin to the point.



b) Draw the right triangle formed by the vector z and the Real axis, and use the Pythagorean Theorem to find the length of the hypotenuse. This also represents the distance from the origin to $1 + 2\mathbf{i}$, or the length of that vector.

c) Use trigonometry to find the angle between z and the Real Axis, and label it on the triangle

d) Consider the multiplication $(1 + 2\mathbf{i}) \cdot 1 = (1 + 2\mathbf{i})$. Plot the vector for 1 on the complex plane above. If we start with 1, and end up at $1 + 2\mathbf{i}$ after multiplication, what factor has the length been scaled by? What angle has the vector been rotated by?

1.3 How to Calculate Stretching and Rotation

We can show that this scale factor and angle are (sort-of) unique to each complex number. That is, each complex number results in a certain stretch and rotation when you multiply by it.

The stretch is equal to the length of the vector representing the complex number (i.e. the distance from the origin). The technical term for this is the **Magnitude**, and we represent it with absolute value signs or $\text{Mag}(z)$. Fittingly, this length is always positive. We can find this using the Pythagorean Theorem on the real and imaginary parts. For $z = a + b\mathbf{i}$:

$$\text{Mag}(z) = |z| = \sqrt{(\text{Re}(z))^2 + (\text{Im}(z))^2} = \sqrt{a^2 + b^2} \tag{8}$$

Or, by multiplying by the complex conjugate z^* (if $z = a + b\mathbf{i}$, then $z^* = a - b\mathbf{i}$)

$$|z|^2 = z \cdot z^* = z^* z \tag{9}$$

The angle of rotation resulting from multiplication is equal to the counterclockwise angle from the Real axis to the vector. This is called the **Argument** or $\text{Arg}(z)$. I will call it θ. We can calculate this using the arctan (inverse tangent) function, but we have to be careful that we are measuring the angle to our vector counterclockwise from the x axis. From now on, this assignment will use radians exclusively.

When we are in the first quadrant (positive Real and Imaginary Parts), arctan works exactly as it did in question 5c. For $z = a + b\mathbf{i}$:

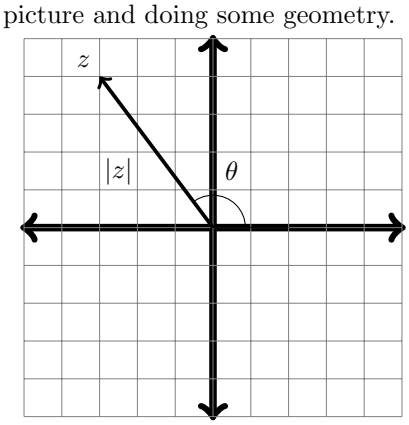
$$\text{Arg}(z) = \theta = \arctan\left(\frac{b}{a}\right), \quad \text{for} \quad a > 0, b > 0 \tag{10}$$

For the other sections of the complex plane, we just have to add on some multiples of π depending on where we are since arctan gives us a related, but incorrect angle.

$$\theta = \arctan\left(\frac{b}{a}\right) + \pi, \quad \text{for} \quad a < 0 \tag{11}$$

$$\theta = \arctan\left(\frac{b}{a}\right) + 2\pi, \quad \text{for} \quad a > 0, b < 0 \tag{12}$$

Don't worry about these equations too much. You should always be able to figure out the correct angle by drawing a picture and doing some geometry.



1.3.1 Problem 6: Let's Calculate Stretch and Rotation

Using the methods from the previous section, calculate the Magnitude $|z|$ and Argument θ for each of the following complex numbers. Draw a picture if you're confused! Hint:
$|z| \geq 0$ (always) and $0 \leq \theta < 2\pi$ (for now)

a) 0 (can we even define θ for this?)

b) 1

c) $\mathbf{i}$

d) -1

e) $-\mathbf{i}$

f) $1 + \mathbf{i}$

g) $3 - 4\mathbf{i}$

h) $-2 + 2\mathbf{i}$

i) $1 - 4\mathbf{i}$

j) $-3 - 6\mathbf{i}$

k) $1 + (\sqrt{3})\mathbf{i}$

1.3.2 Problem 7: Going the Other Direction

Use trigonometry (draw a picture if you need to!) to find a complex number $z = a + b\mathbf{i}$ that would correspond to each pair of stretch and rotation

a) $|z| = 6$, $\theta = 0$

b) $|z| = 1$, $\theta = \frac{\pi}{2}$

c) $|z| = 2$, $\theta = \frac{3\pi}{4}$

d) $|z| = 1$, $\theta = \frac{13\pi}{6}$

e) $|z| = 1$, $\theta = \pi$

2 Building The Complex Exponential

2.1 Why Exponentials? A Review of Exponent Laws

Normally when we represent rotation, we are adding angles together. For example, a rotation by $\frac{\pi}{2}$ radians plus another rotation of $\frac{\pi}{2}$ radians equals π radians in total. In general, any combination of rotations can be represented as a sum of angles:

$$\theta_{\text{total}} = \theta_1 + \theta_2 \tag{13}$$

But, complex numbers accumulate rotation when we multiply them! If multiplying by z_1 results in a rotation of θ_1 and z_2 results in a rotation of θ_2, then their product $z_1 \cdot z_2$ will result in a rotation of $\theta_1 + \theta_2$. So in order to describe this behavior, we need something that creates sums when we do multiplication. In the language of complex numbers:

$$\text{Arg}(z_1 \cdot z_2) = \text{Arg}(z_1) + \text{Arg}(z_2) \tag{14}$$

Which looks suspiciously like the sum law for logarithms

2.1.1 Exponent Laws

Let's review the rules for combining exponents.

$$b^m b^n = b^{m+n} \tag{15}$$

$$\frac{b^m}{b^n} = b^m b^{-n} = b^{m-n} \tag{16}$$

$$(b^m)^n = b^{m \cdot n} \tag{17}$$

Equation 15 is promising, because we are multiplying two quantities (b^m and b^n), but we end up calculating a sum (the $m + n$ in the exponent). This gives us a hint that if we somehow store our rotational angle information in the exponent, then we might be able to replicate the behavior we saw in complex numbers. That is, representing rotation (a fundamentally additive process) with multiplication.

2.2 What happens if we put i in the exponent?

There is an issue with expressing rotations with exponential functions. Exponential functions (Real ones at least) do not look like rotation at all! 2^x zooms away from the x-axis at breakneck speed, but pure rotations (no stretching) stay the same distance from the origin, and go at a constant speed. Perhaps in our desperation to find an exponential representation for rotation, we come up with the (seemingly nonsensical) expression:

$$A^{\mathbf{i}} \tag{18}$$

where A is a Real Number. This turns out to be exactly what we need. We can show that it is equivalent to a complex number that represents a pure rotation. This idea occurs

more naturally from the viewpoint of calculus (specifically infinite series), but we can explore an imaginary exponent with the tools at hand.

2.2.1 Problem 8: Proving $A^{\mathbf{i}}$ is a pure rotation

To show that $A^{\mathbf{i}}$ is a pure rotation, we need to show that it causes no stretching. That is, its magnitude $|A|$ is 1. Remember, A is a placeholder variable that can represent **any** Real Number.

a) We can represent $A^{\mathbf{i}}$ as a complex number with real and imaginary parts: $A^{\mathbf{i}} = x + \mathbf{i}y$

Use Equation 8 to find the magnitude of $A^{\mathbf{i}}$ in terms of x and y

b) Find $(A^{\mathbf{i}})^*$=$A^{-\mathbf{i}}$, the complex conjugate of $A^{\mathbf{i}}$ in terms of x and y

c) use Equation 9 to find the magnitude of $A^{\mathbf{i}}$ in terms of x and y (Use FOIL!) You should get the same answer as before.

d) use Equation 15 to simplify $A^{\mathbf{i}} \cdot A^{-\mathbf{i}}$ (This is the magnitude of $A^{\mathbf{i}}$ squared!)

e) set $|A^{\mathbf{i}}|^2$ in terms of x and y and your answer to part d equal to each other. What is this the equation of?

In the previous problem, we proved that any Real Number taken to an Imaginary power is a complex number with magnitude 1. In part e, we saw that these complex numbers lie on the unit circle in the complex plane. In fact, the angle of rotation due to multiplication by $A^{\mathbf{i}}$ is equal to the angle of its position on the unit circle.
Let's investigate what happens when we do successive rotations. Remember, whatever is in the exponent represents the amount of rotation we're doing. Multiplying by $A^{\mathbf{i}}$ once will rotate us by some amount θ. Since $A^{\mathbf{i}} = A^{\mathbf{i}\cdot 1}$, we can say that $\theta = 1$. This isn't in degrees or radians, the units of this angle depend on the A we choose. Multiplying by $A^{\mathbf{i}}$ twice will give us twice the amount of rotation:

$$A^{\mathbf{i}} \cdot A^{\mathbf{i}} = A^{\mathbf{i}+\mathbf{i}} = A^{2\mathbf{i}} \tag{19}$$

The exponent is twice as large, so we are rotating twice as much. We have sucessfully created an exponential object where additive rotation can be represented by multiplication. Importantly, since $2\mathbf{i}$ is still imaginary, our proof in problem 8 still applies and $|A^{2\mathbf{i}}| = 1$. This means the complex exponential is staying the same distance from the origin.

2.3 Euler's Constant e

$e \approx 2.71828$ is a fundamental mathematical constant like π. It has a few precise definitions, but they all use calculus:
As a limit (what this function approaches as n goes to infinity):

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \tag{20}$$

As an infinite sum:

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} = 1 + \frac{1}{1} + \frac{1}{1\cdot 2} + \frac{1}{1\cdot 2\cdot 3} + \frac{1}{1\cdot 2\cdot 3\cdot 4} + ... \tag{21}$$

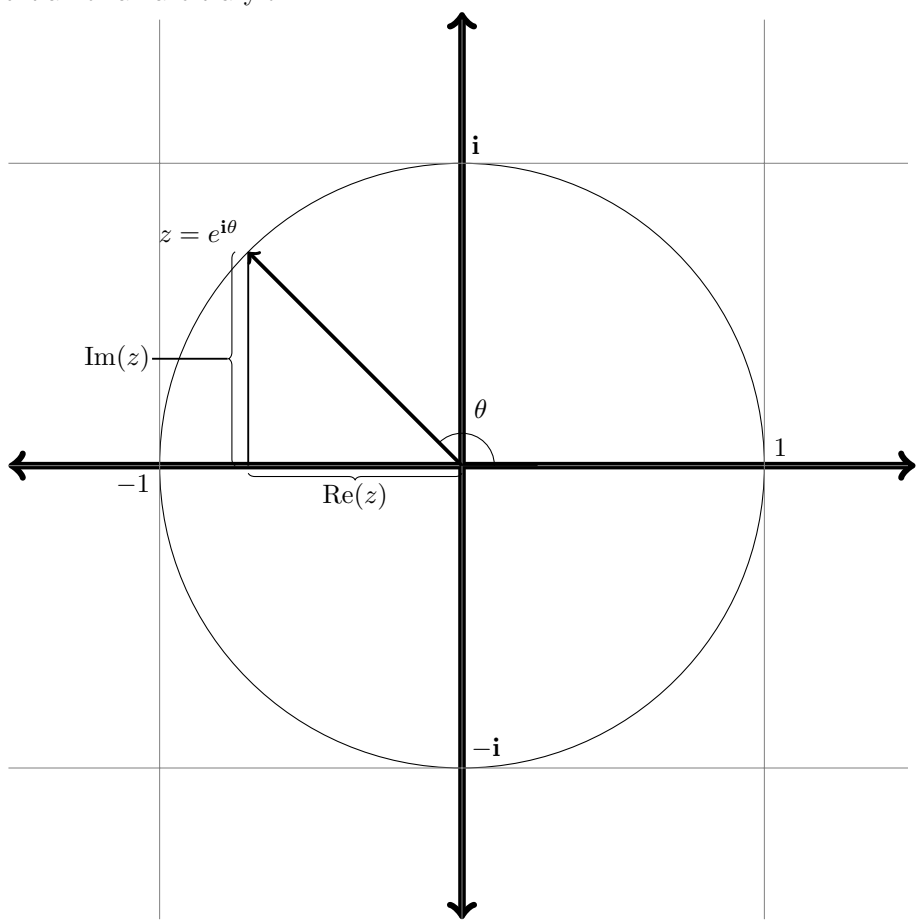
Don't worry too much about these, we will not be using them. The rest of this section will primarily be conceptual, since the results come from calculus. Luckily, we will be able to use the unique properties of e without understanding exactly where they come from. The main property of e that we care about is that when we use it to encode rotational information about an exponential in $e^{\mathbf{i}\theta}$, θ is measured in radians. That is, $\theta = 2\pi$ rotates us around the unit circle **exactly once**, leaving us right where we started.

$$e^{\mathbf{i}(2\pi)} = 1 \tag{22}$$

Using a smaller number (for example $2^{\mathbf{i}\theta}$) will make the rotations too small, and a larger one will make the rotations too large (for example $10^{\mathbf{i}\theta}$).

2.4 Euler's Formula

Because $e^{\mathbf{i}\theta}$ creates angles measured in radians, we can come up with a formula that gives us the real an imaginary parts of $e^{\mathbf{i}\theta}$ for any θ. Let's investigate the picture of a complex exponential for an arbitrary θ:



This looks very similar to the unit circle we created on the x-y plane (see **Real Unit Circle** in Appendix). In fact, we can compare the two, and come to the following conclusions: The Real part of $e^{\mathbf{i}\theta}$ is $\cos(\theta)$ and the Imaginary part is $\sin(\theta)$. Putting this together gives us Euler's Formula.

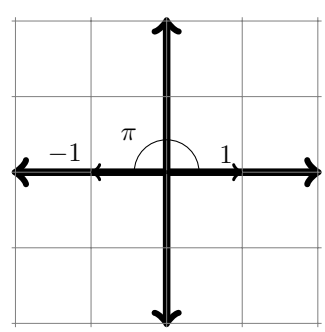
Principle 2.1. Euler's Formula:

$$e^{\mathbf{i}\theta} = \cos(\theta) + \mathbf{i}\sin(\theta) \tag{23}$$

Plugging in π for θ gives us the famous equation:

$$e^{\mathbf{i}\pi} = -1 \tag{24}$$

which says that a rotation of π radians (180 degrees) in the complex plane is equivalent to multiplying by -1



Euler's formula is one of the most beautiful results in mathematics. It relates exponential functions to trig functions, which at first glance seem totally different. It shows an intimate relationship between two fundamental constants π and e. And it turns out to be everywhere in math and science. To understand Euler's formula conceptually, just remember that $e^{\mathbf{i}\theta}$ is a rotation in complex plane counterclockwise from the Real axis measured in radians.

2.4.1 Reverse Euler's Formulas

We can also reverse Euler's formula to express sine and cosine in terms of complex exponentials:

Principle 2.2. Reverse Euler's Formulas:

$$\cos(\theta) = \frac{e^{\mathbf{i}\theta} + e^{-\mathbf{i}\theta}}{2} \tag{25}$$

$$\sin(\theta) = \frac{e^{\mathbf{i}\theta} - e^{-\mathbf{i}\theta}}{2\mathbf{i}} \tag{26}$$

2.4.2 Problem 9: Using Euler's Formula

Any formula no matter how beautiful is pretty useless if we can't actually use it. Express each complex number below in $a + bi$ form. Keep radicals from special right triangles if you can. Otherwise, round to the nearest hundredth. Use Euler's formula to find the Real and Imaginary Parts. (make sure your calculator is in radian mode!)

a) $e^{\mathbf{i}}$

b) $e^{\mathbf{i}\frac{\pi}{2}}$

c) $e^{\mathbf{i}\frac{7\pi}{6}}$

d) $4(e^{10.79\mathbf{i}})$

3 Applications of The Complex Exponential

3.1 Trig Identities

Trigonometric Identities are formulas that relate the trig functions to one another. The most well known may be the Pythagorean Identity:

$$\sin^2(\theta) + \cos^2(\theta) = 1 \tag{27}$$

This can be proven by applying the Pythagorean Theorem to the right triangle in the Unit Circle (see appendix). We can also prove this with Euler's Formula. Consider the magnitude of a complex exponential. From Equation 9,

$$|e^{\mathbf{i}\theta}|^2 = e^{\mathbf{i}\theta} \cdot (e^{\mathbf{i}\theta})^* = e^{\mathbf{i}\theta} \cdot (e^{-\mathbf{i}\theta}) = e^{\mathbf{i}(\theta-\theta)} = e^0 = 1 \tag{28}$$

and from Euler's Formula:

$$|e^{\mathbf{i}\theta}|^2 = e^{\mathbf{i}\theta} \cdot (e^{\mathbf{i}\theta})^* = (\cos(\theta) + \mathbf{i}\sin(\theta)) \cdot (\cos(\theta) - \mathbf{i}\sin(\theta)) = \cos^2(\theta) + \sin^2(\theta) \tag{29}$$

Setting Equation 28 equal to Equation 29 gives us the Pythagorean Identity:

$$\sin^2(\theta) + \cos^2(\theta) = 1 \tag{30}$$

3.1.1 Problem 10: Proving Trig Identities With Euler's Formula

As you continue your mathematical career, you will encounter many trig identities. All of them can be proven with Euler's Formula. The general recipe for this type of proof goes like this:

1. Recognize that one side of the trig identity is the Real/Imaginary part of some complex exponential
2. Use exponent laws to manipulate that complex exponential (see Section 2.1.1 for exponent laws), then use Euler's Formula to convert exponentials to trig
3. Take the Real/Imaginary part to get to the desired trig expression (note: you must use the same part for the first and last step i.e. real and real or imaginary and imaginary)

Another strategy that may also work:

1. Express both sides of the identity using the Reverse Euler's Formulas
2. Use exponent laws to show they are equivalent

These will not always be so straightforward! However, the proof based on Euler's Formula is often much easier than a purely geometric proof. With that in mind, use Euler's formula to prove each of the following identities:

a) The Phase Shift Identity:

$$\sin(\frac{\pi}{2} - \theta) = \cos\theta \tag{31}$$

b) The Double Angle Identities:

$$\sin(2\theta) = 2\sin(\theta)\cos(\theta) \tag{32}$$

$$\cos(2\theta) = 2\cos^2(\theta) - \sin^2(\theta) \tag{33}$$

c) The Angle Sum Identities:

$$\sin(\theta + \phi) = \sin(\theta)\cos(\phi) + \cos(\theta)\sin(\phi) \tag{34}$$

$$\cos(\theta + \phi) = \cos(\theta)\cos(\phi) - \sin(\theta)\sin(\phi) \tag{35}$$

d) The Reverse Euler's Formulas (Equations 25 and 26)

3.2 A Better Representation for Complex Numbers

In section 1.2 we learned that multiplying by any complex number can be represented as a stretch combined with a rotation. We can use this idea along with complex exponentials to create a new way to write complex numbers that makes multiplication easier.

Principle 3.1. Any complex number $z = a + b\mathbf{i}$ can be written in the form:

$$z = re^{\mathbf{i}\theta} \tag{36}$$

where r is the distance from the origin (the magnitude of z):

$$r = \text{Mag}(z) = |z| = \sqrt{z \cdot z^*} = \sqrt{a^2 + b^2} \tag{37}$$

And θ is the angle of rotation (the argument of z, see Equations 10-12):

$$\theta = \text{Arg}(z) \tag{38}$$

r causes the stretch, and $e^{\mathbf{i}\theta}$ causes the rotation.
We can also go from the complex exponential form to the $a + b\mathbf{i}$ form:

$$a = \text{Re}(z) = r\cos(\theta) \tag{39}$$

$$b = \text{Im}(z) = r\sin(\theta) \tag{40}$$

If we want to multiply 2 complex numbers $z_1 = r_1 e^{\mathbf{i}\theta_1}$ and $z_2 = r_2 e^{\mathbf{i}\theta_2}$
We have:

$$z_1 \cdot z_2 = (r_1 e^{\mathbf{i}\theta_1}) \cdot r_2 e^{\mathbf{i}\theta_2} = r_1 r_2 e^{\mathbf{i}\theta_1} e^{\mathbf{i}\theta_2} = r_1 r_2 e^{\mathbf{i}(\theta_1 + \theta_2)} \tag{41}$$

(multiply the magnitudes and add the angles)

3.3 Rotational Motion

In Physics, it is often advantageous to express rotational motion in terms of complex exponentials instead of sine and cosine. The complex exponential contains all the same information (and more), but as you've seen it is easier to work with because we can leverage the power of exponent laws. This motion can be visualized with the image on the front of this packet. We can define a function:

$$p(t) = re^{\mathbf{i}\omega t} \tag{42}$$

This represents a point traveling around a circle of radius r in the complex plane traveling at an angular frequency of ω (radians/second). Notice that the real and imaginary parts give us cosine and sine:

$$\text{Re}(p(t)) = r\cos(\omega t) \tag{43}$$

$$\text{Im}(p(t)) = r\sin(\omega t) \tag{44}$$

Calculus can be used to find the velocity and acceleration of this point. (don't worry about how this is done, we will just interpret the results) The velocity $v(t)$:

$$v(t) = \mathbf{i}\omega r e^{\mathbf{i}\omega t} \tag{45}$$

The magnitude of the velocity (speed) is:

$$|v| = r\omega \tag{46}$$

so the point is traveling at a constant speed while moving in a circle. Also notice that in Equation 45, the velocity is equal to $\mathbf{i}\omega$ times the position:

$$v(t) = \mathbf{i}\omega \cdot p(t) \tag{47}$$

So, the velocity is rotated by $\frac{\pi}{2}$ radians (90°) relative to the position, and stretched by ω. The position vector is always a radius of the circular path of motion. Since the velocity is perpendicular to the radius, it is tangent to the circle (if drawn as an arrow away from the position).
The acceleration is:

$$a(t) = \mathbf{i}^2\omega^2 r e^{\mathbf{i}\omega t} = -\omega^2 r e^{\mathbf{i}\omega t} \tag{48}$$

The magnitude of the acceleration is:

$$|a| = \omega^2 r \tag{49}$$

So, if the point has a mass m, then a force of magnitude $mr\omega^2$ must be acting on it to create the circular motion.
In Equation 48, we can see that the acceleration is equal to $-\omega^2$ times the position:

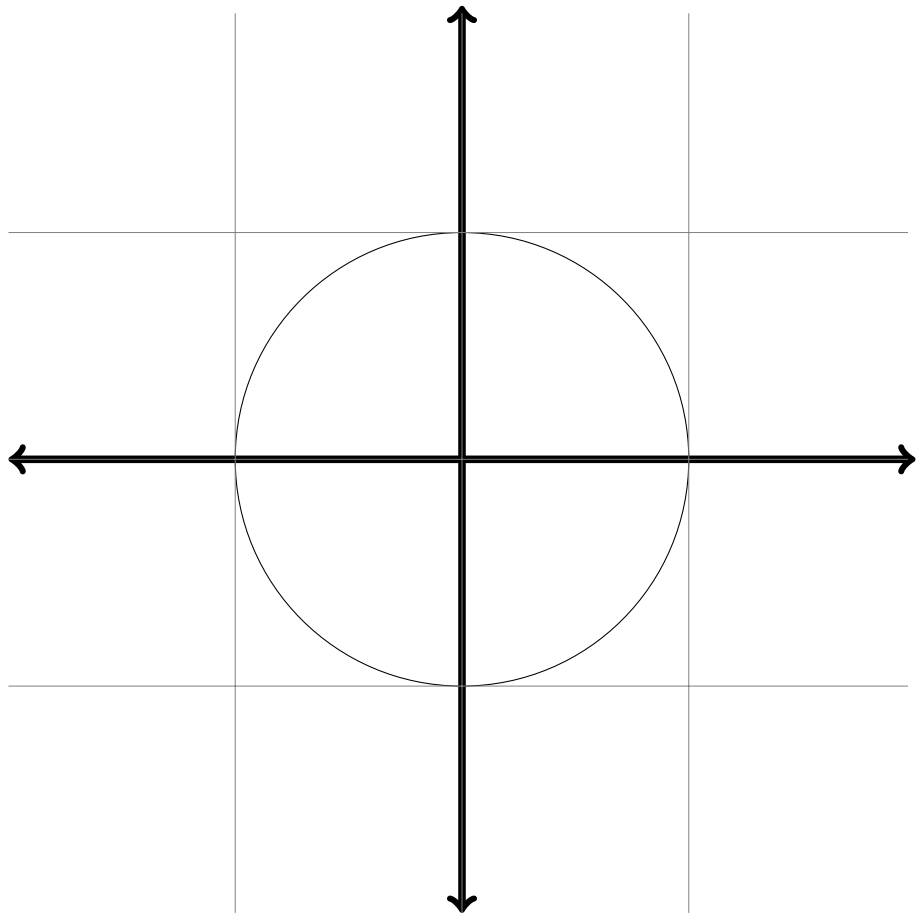
$$a(t) = -\omega^2 \cdot p(t) \tag{50}$$

So, the acceleration points the opposite direction as the position (rotated by π radians), and it is stretched by ω^2. The acceleration points towards the center of the circle.

3.3.1 Problem 11: Rotational Motion Diagram

Use the information in section 3.3 to draw a diagram of a particle traveling in a circular path. Its position is defined by the function $p(t) = e^{\mathbf{i}t}$. One the next page, follow these steps:

1. plot the position at $t = \frac{\pi}{4}$ and $t = \frac{3\pi}{4}$ as point on the unit circle.
2. find the velocity at $t = \frac{\pi}{4}$ and $t = \frac{3\pi}{4}$ and draw it as an arrow from the position point.
3. find the acceleration at $t = \frac{\pi}{4}$ and $t = \frac{3\pi}{4}$ and draw it as an arrow from the position point.



a) how long does it take for the particle to travel around the circle once? (assuming t is measured in seconds)

b) how do we know the velocity is perpendicular to the position vector?

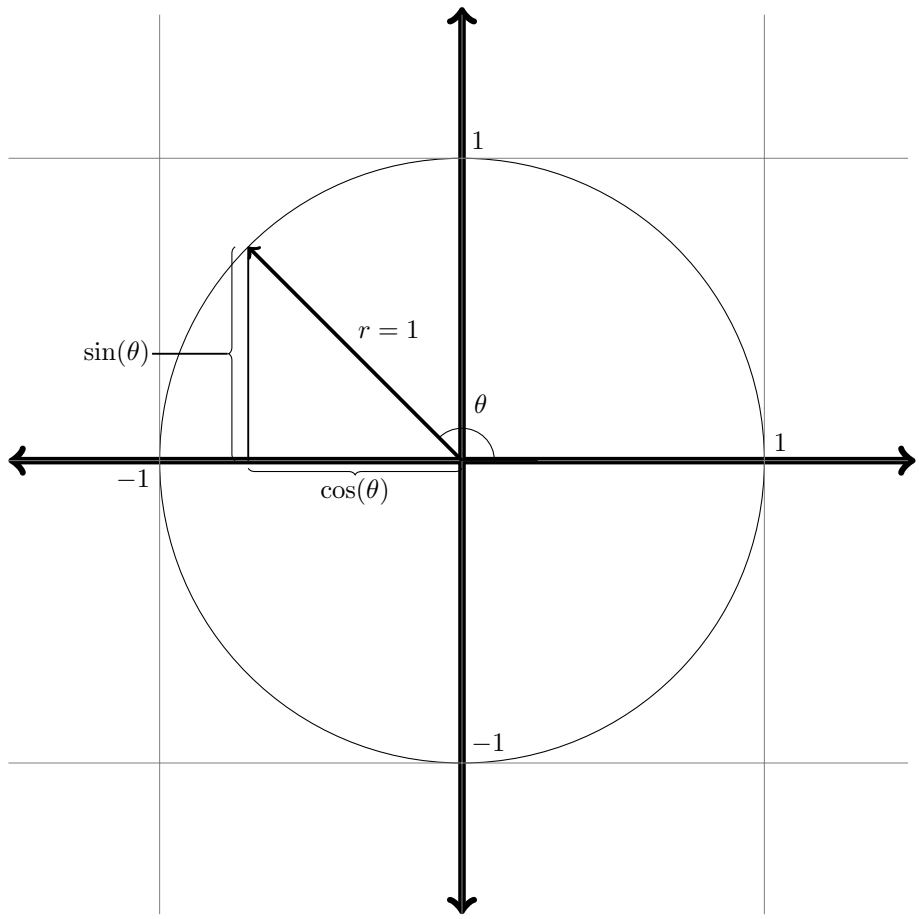
c) what is the speed of the particle (assuming r is measured in meters)?

4 Conclusion/Reflection

Hopefully this assignment has given you some idea of what the complex exponential is and how it's useful. Please write a short reflection about this assignment. Feel free to ask any questions you have about complex numbers, the complex exponential or math in general, and I will attempt to answer. Some things you may think about: Can you think of any other places you might use a complex exponential? Do you think complex numbers are a real thing, or just a construct? Are ideas any less real than physical objects? Are the complex numbers any less real than Real numbers or counting numbers?

5 Appendix

5.1 Real Unit Circle



References

[1] Tristan Needham (1997) *Visual Complex Analysis*, Clarendon Press.

[2] Matt Calhoun (2010-11-10) *Understanding Imaginary Exponents*, Math Stack Exchange, https://math.stackexchange.com/q/9783

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